

Practical-7

7. Write a program that prints the addresses of code, global, static, heap, and stack variables. Analyze the Linux process address space layout and explain each memory segment.

Program:

```
#include <stdio.h>
#include <stdlib.h>
int global=100;
static int s=50;

int main()
{
    int stack=10;
    int *heap=(int*)malloc(sizeof(int));

    printf("Code Segment : %p\n", main);
    printf("Global Segment : %p\n", &global);
    printf("Static Segment : %p\n", &s);
    printf("Heap Segment : %p\n", heap);
    printf("Stack Segment : %p\n", &stack);
    free(heap);
    return 0;
}
```

Make use of the /proc/maps file and memory analysis tools to study the memory organization of a running process. Prepare a report explaining virtual memory mappings.

```
GNU nano 8.7.1 practical7.c *
#include <stdio.h>
#include <stdlib.h>

int global = 100;
static int s = 50;

int main()
{
    int stack = 10;
    int *heap = (int*)malloc(sizeof(int));

    printf("Code Segment : %p\n", main);
    printf("Global Segment : %p\n", &global);
    printf("Static Segment : %p\n", &s);
    printf("Heap Segment : %p\n", heap);
    printf("Stack Segment : %p\n", &stack);

    free(heap);
    return 0;
}
```

```
mokshitha_putti@Mokshithaf x + v - □ X
mokshitha_putti@MokshithaPutti:~$ mkdir -p ~/OSLAB/practicals
mokshitha_putti@MokshithaPutti:~$ cd ~/OSLAB/practicals
mokshitha_putti@MokshithaPutti:~/OSLAB/practicals$ pwd
/home/mokshitha_putti/OSLAB/practicals
mokshitha_putti@MokshithaPutti:~/OSLAB/practicals$ nano practical7.c
mokshitha_putti@MokshithaPutti:~/OSLAB/practicals$ gcc practical7.c -o practical7
mokshitha_putti@MokshithaPutti:~/OSLAB/practicals$ ls
practical7 practical7.c
mokshitha_putti@MokshithaPutti:~/OSLAB/practicals$ ./practical7
Code Segment : 0x5e3c4ae3d1a9
Global Segment : 0x5e3c4ae40010
Static Segment : 0x5e3c4ae40014
Heap Segment : 0x5e3c662fa010
Stack Segment : 0x7ffcb31138bc
mokshitha_putti@MokshithaPutti:~/OSLAB/practicals$ ps
  PID TTY          TIME CMD
  336 pts/0        00:00:00 bash
  574 pts/0        00:00:00 ps
mokshitha_putti@MokshithaPutti:~/OSLAB/practicals$ cat /proc/336/maps
5ac5a180c000-5ac5a183d000 r--p 00000000 08:30 1502 /us
r/bin/bash
5ac5a183d000-5ac5a1940000 r-xp 00031000 08:30 1502 /us
r/bin/bash
5ac5a1940000-5ac5a1977000 r--p 00134000 08:30 1502 /us
r/bin/bash
5ac5a1977000-5ac5a197b000 r--p 0016b000 08:30 1502 /us
r/bin/bash
5ac5a197b000-5ac5a1984000 rw-p 0016f000 08:30 1502 /us
r/bin/bash
5ac5a1984000-5ac5a1990000 rw-p 00000000 00:00 0
5ac5c6823000-5ac5c698e000 rw-p 00000000 00:00 0 [he
ap]
78e150000000-78e150028000 r--p 00000000 08:30 13641 /us
r/lib/x86_64-linux-gnu/libc.so.6
78e150028000-78e1501c0000 r-xp 00028000 08:30 13641 /us
r/lib/x86_64-linux-gnu/libc.so.6
78e1501c0000-78e15020e000 r--p 001c0000 08:30 13641 /us
r/lib/x86_64-linux-gnu/libc.so.6
78e15020e000-78e150212000 r--p 0020d000 08:30 13641 /us
r/lib/x86_64-linux-gnu/libc.so.6
```

```

mokshitha_putti@MokshithaF x + v - □ x
78e150400000-78e15040e000 r--p 00000000 08:30 14513 /us
r/lib/x86_64-linux-gnu/libtinfo.so.6.6
78e15040e000-78e150422000 r-xp 0000e000 08:30 14513 /us
r/lib/x86_64-linux-gnu/libtinfo.so.6.6
78e150422000-78e150430000 r--p 00022000 08:30 14513 /us
r/lib/x86_64-linux-gnu/libtinfo.so.6.6
78e150430000-78e150434000 r--p 0002f000 08:30 14513 /us
r/lib/x86_64-linux-gnu/libtinfo.so.6.6
78e150434000-78e150435000 rw-p 00033000 08:30 14513 /us
r/lib/x86_64-linux-gnu/libtinfo.so.6.6
78e150435000-78e150436000 r--p 00000000 08:30 13690 /us
r/lib/locale/C.utf8/LC_PAPER
78e150436000-78e150437000 r--p 00000000 08:30 13688 /us
r/lib/locale/C.utf8/LC_NAME
78e150437000-78e150438000 r--p 00000000 08:30 13681 /us
r/lib/locale/C.utf8/LC_ADDRESS
78e150438000-78e150439000 r--p 00000000 08:30 13691 /us
r/lib/locale/C.utf8/LC_TELEPHONE
78e150439000-78e15043a000 r--p 00000000 08:30 13685 /us
r/lib/locale/C.utf8/LC_MEASUREMENT
78e15043a000-78e15043b000 r--p 00000000 08:30 13684 /us
r/lib/locale/C.utf8/LC_IDENTIFICATION
78e15043b000-78e15043d000 rw-p 00000000 00:00 0
78e15043d000-78e15043e000 r--p 00000000 08:30 13638 /us
r/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
78e15043e000-78e15046d000 r-xp 00001000 08:30 13638 /us
r/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
78e15046d000-78e150478000 r--p 00030000 08:30 13638 /us
r/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
78e150478000-78e15047a000 r--p 0003b000 08:30 13638 /us
r/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
78e15047a000-78e15047b000 rw-p 0003d000 08:30 13638 /us
r/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
78e15047b000-78e15047c000 rw-p 00000000 00:00 0
7fff4d516000-7fff4d537000 rw-p 00000000 00:00 0 [st
ack]
7fff4d53c000-7fff4d540000 r--p 00000000 00:00 0 [vv
ar]
7fff4d540000-7fff4d542000 r-xp 00000000 00:00 0 [vd
so]
mokshitha_putti@MokshithaPutti:~/OSLAB/practicals$ |

```

Practical-8a

8. Develop a program using `malloc()`, `calloc()`, `realloc()`, and `free()`. Monitor memory allocation behavior and identify memory leaks using Valgrind.

Program:

```
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int *a,*b;

    a=(int*)malloc(5*sizeof(int));

    b=(int*)calloc(5,sizeof(int));

    a=(int*)realloc(a,10*sizeof(int));

    free(a);
    free(b);

    return 0;
}
```

Practical -8b

Run valgrind and check the result as: All heap blocks were freed -- no leaks are possible.

Develop a program that demonstrates Copy-on-Write behavior after a `fork()`. Analyze memory usage before and after modifying data in the child process.

```
#include <stdio.h>
#include <unistd.h>
int main()
{
    int x=100;
    if(fork()==0)
    {
        printf("Child Before=%d\n",x);
        x=500;
        printf("Child After=%d\n",x);
    }
    else
    {
        sleep(2);
        printf("Parent=%d\n",x);
    }

    return 0;
}
```

```
mokshitha_putti@MokshithaPutti:~/OSLAB/practicals$ nano practical8a.c
mokshitha_putti@MokshithaPutti:~/OSLAB/practicals$ gcc -g practical8a.c -o practical8a
mokshitha_putti@MokshithaPutti:~/OSLAB/practicals$ ./practical8a
mokshitha_putti@MokshithaPutti:~/OSLAB/practicals$ valgrind --version
valgrind-3.26.0
mokshitha_putti@MokshithaPutti:~/OSLAB/practicals$ sudo apt update
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [543 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:6 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [679 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [127 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [162 kB]
Get:9 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.7 kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [98.0 kB]
Get:11 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [187 kB]
Get:12 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [308 kB]
Get:13 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [59.3 kB]
Get:14 http://archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [98.9 kB]
Get:15 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.4 kB]
Get:16 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [443 kB]
Get:17 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Comp
```